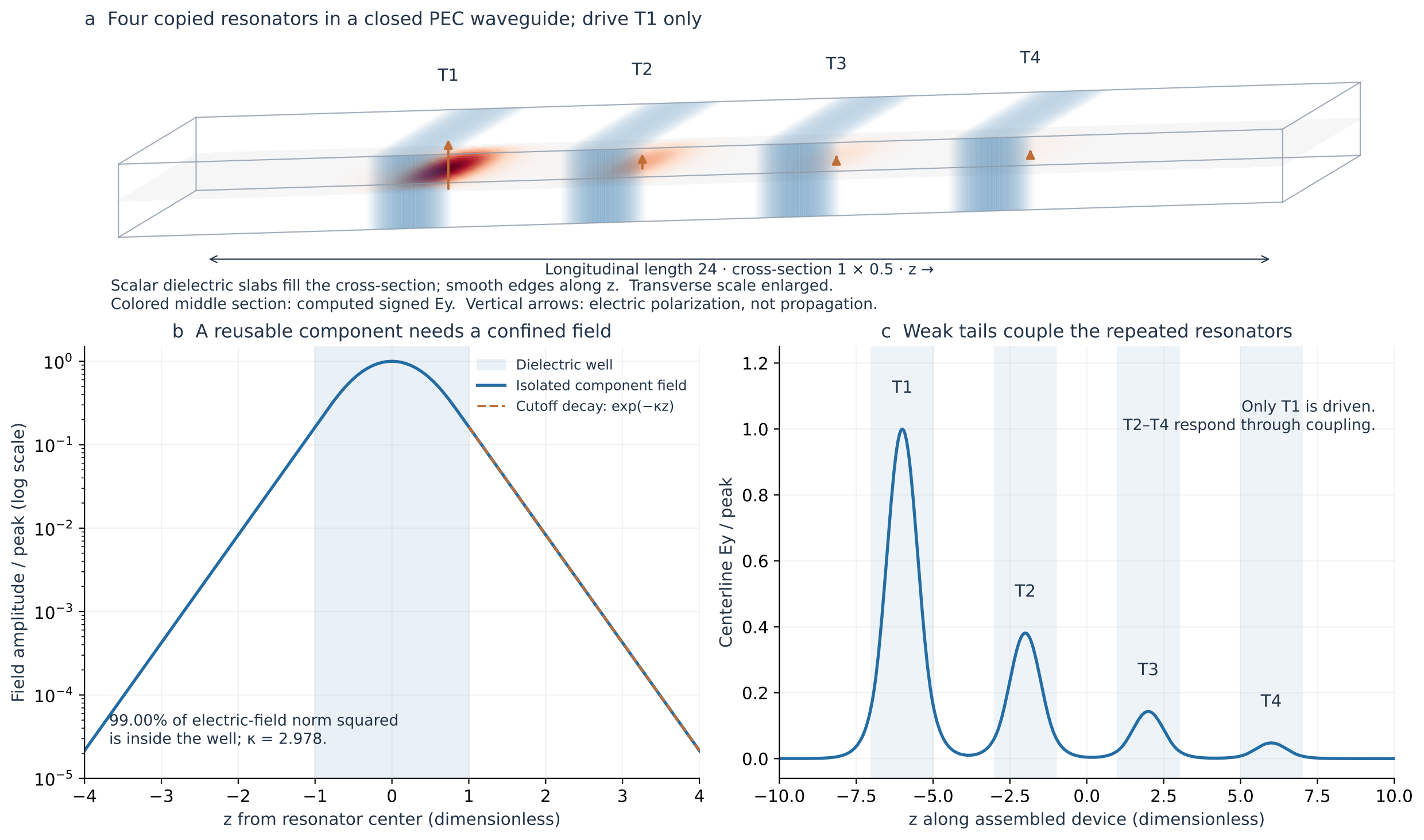

a Four copied resonators in a closed PEC waveguide; drive T1 only
T1
T2
T3
T4
Longitudinal length 24 · cross-section 1 × 0.5 · z →
Scalar dielectric slabs fill the cross-section; smooth edges along z. Transverse scale enlarged.
Colored middle section: computed signed Ey. Vertical arrows: electric polarization, not propagation.
b A reusable component needs a confined field
Dielectric well
Isolated component field
Cutoff decay: exp(−κz)
Field amplitude / peak (log scale)
99.00% of electric-field norm squared
is inside the well; κ = 2.978.
z from resonator center (dimensionless)
c Weak tails couple the repeated resonators
Only T1 is driven.
T2–T4 respond through coupling.
Centerline Ey / peak
z along assembled device (dimensionless)